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Machine learning for water quality management: a systematic review of prediction models, data challenges, and future directions

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Abstract

Water quality management faces major challenges due to nonlinearity, data scarcity, spatial heterogeneity, and monitoring costs. Although numerous studies address machine learning (ML), there is a lack of comprehensive research on how ML practically overcomes these challenges. To fill this gap, this review categorizes ML capabilities into functional themes. Following PRISMA guidelines, this study synthesizes 86 peer-reviewed articles (2014–2026), prioritizing specific operational gains over general accuracy claims. For instance, random forest reduced RMSE by 60.1% compared to linear models, improving prediction accuracy. Moreover, ML models handle 50–70% missing data in time series reconstruction. To address data scarcity, transfer learning leverages knowledge from data-rich regions to enhance prediction efficiency (NSE) in data-poor regions by 7–17%. Unsupervised ML also excels at identifying hidden pollution sources through chemical fingerprinting, without prior labeling. Concurrently, feature ranking significantly lowers monitoring costs by reducing necessary variables from 13 to just 4, while maintaining robust prediction accuracy and enhancing model generalizability. Building upon these capabilities, integrating ML with IoT frameworks enables highly effective early-warning management, achieving up to 99.8% accuracy in real-time anomaly detection. Despite these advancements, the review carefully addresses technical risks such as overfitting and overestimation, emphasizing that continuous human oversight remains essential. The findings also revealed that no one-size model fits all scenarios, as predictive performance depends on data nature, sample size, model architecture, and environmental conditions. Ultimately, ML enables smart water-quality management, representing a qualitative shift toward proactive environmental resilience.

Highlights

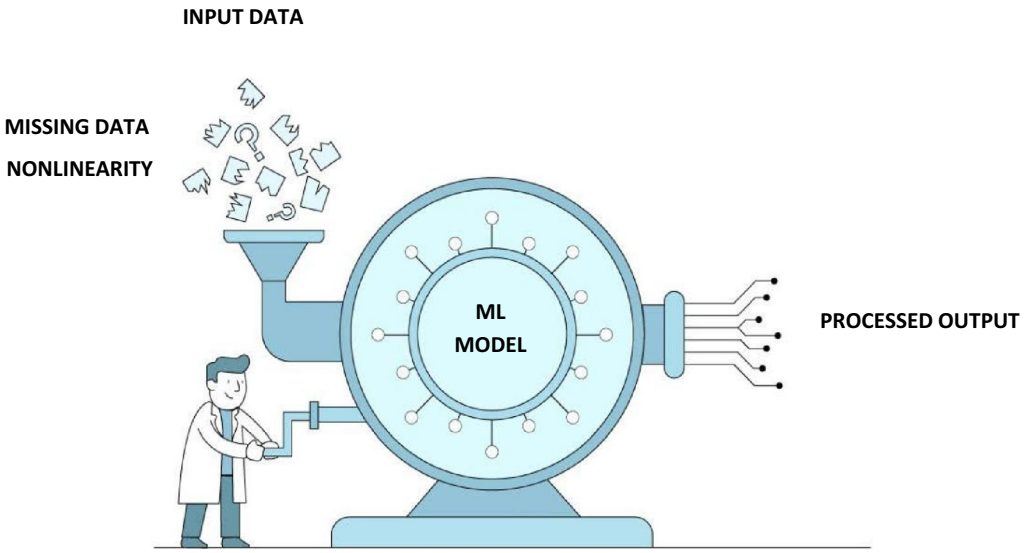
- ML successfully handles 50–70% of missing data in time series.
- ML reduces the error rate by 60.1% compared to traditional models.
- These models excel at feature ranking and data dimensionality reduction.
- Prioritizing variables enables cost-effective and real-time monitoring.
- Integration with IoT achieved up to 99.8% anomaly detection accuracy.

Keywords Machine learning, Artificial intelligence, Water quality prediction, Sustainable development, Water quality index

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Graphical abstract



Introduction

According to [1], ensuring the availability and sustainable management of water and sanitation for all is a core target of Sustainable Development Goal 6 (SDG 6). Achieving this goal requires robust monitoring systems. However, water quality management poses a significant challenge in environmental and hydrogeological studies. This is due to the complexity of nonlinear relationships between physical and chemical parameters, spatial variability, and data scarcity. In this context, machine learning techniques (ML) have provided a promising framework for addressing these challenges. They contribute to improving prediction accuracy through advanced modeling and data-driven decision-making [2, 3]. Machine learning models are characterized by their superior ability to process high-dimensional data and extract complex nonlinear relationships. This improves model generalizability and reduces the need for rigid statistical assumptions compared to traditional methods.

Furthermore, feature importance analysis ranks variable impacts, enabling a focus on critical factors. This facilitates management, reduces monitoring costs, and improves decision-making efficiency for water resources [4–6]. These capabilities allow machine learning models to detect significant variables that traditional linear models might overlook. This is due to their high efficiency in representing complex nonlinear relationships and detecting hidden patterns [7]. This review emphasizes that the transition from traditional to ML-based modeling is not only a technical upgrade but a functional necessity for complex hydrogeological systems.

Numerous applied studies have demonstrated the efficiency of machine learning in predicting water quality indicators in different hydrogeological environments. This has included specific parameters such as TDS, nitrates, and heavy metals, as well as composite indicators (WQI). Compared to traditional methods, these models achieve higher accuracy and require fewer computational assumptions. For example, deep learning models have been successfully used to model dynamic temporal changes in water quality. Long-memory neural network (LSTM) models have improved the prediction of quality fluctuations across time series [8, 9]. In Egypt's Nile Delta, AI-based salinity prediction models have enabled effective monitoring of groundwater degradation risks [10].

Additionally, machine learning demonstrates high flexibility in handling incomplete or low-quality data. It processes missing or anomalous values and reconstructs underlying patterns, improving interpretation accuracy [11, 12]. These models uncover hidden patterns, such as chemical fingerprints of pollution sources, enhancing accurate tracking. This technology provides reliable spatial and temporal predictions, enabling sustainable water management strategies [13–16].

Hybrid frameworks, integrating machine learning with traditional hydrochemical and geostatistical methods, significantly enhance prediction accuracy. Advanced techniques like transfer learning overcome data scarcity

by reusing models pre-trained on richer datasets [17, 18]. This represents a significant advantage over traditional statistical methods that rely primarily on local sample size [19].

Emerging machine learning models, such as non-negative matrix analysis (NMF), have provided effective tools for analyzing the chemical composition of groundwater. This technique separates pollution sources, such as fertilizers and waste, and estimates their fluxes without prior knowledge of their composition [20]. This gives it a clear advantage over physical or inverse models, which often require complex parameters or in-depth specialized knowledge.

The rapid development of Internet of Things technologies has enabled the collection and transmission of water data in real time, facilitating the activation of early warning systems and supporting rapid decision-making. As a result, it has become possible to accurately identify high-risk areas and implement necessary preventative measures immediately [14, 21, 22]. This integration of the Internet of Things (IoT) and Machine Learning (ML) is promising in real-time monitoring systems.

Despite their efficiency, machine learning models can overestimate or underestimate actual water quality values. Overfitting is another common problem affecting the accuracy of results. In this case, the model may perform exceptionally well during training but fail to make accurate predictions when tested on new data. This necessitates careful attention to model selection, tuning, and validation. In this context, human expertise becomes crucial in data processing stages, such as cleaning and normalization. This expertise plays a vital role in improving the accuracy and reliability of models, something that automation cannot fully cover. Ying [23] outlined several strategies to mitigate overfitting problem. These solutions include stopping training early, reducing the complexity of the neural network, expanding the dataset, and carefully selecting input features. Tran et al. [24] reported that, the model accuracy depends on the chosen algorithm, specific prediction goals, and input variables. Therefore, selecting the appropriate machine learning model requires a careful evaluation of both the method and the influencing factors.

Unlike previous studies that focused primarily on algorithm structure (e.g., [5, 76]), this study adopts a functional problem-solving perspective to fill a clear research gap. We classify machine learning applications according to their operational gains, offering a practical framework for how these models address specific challenges such as data scarcity, non-linearity, spatial variability, and high monitoring costs (Fig. 1). Furthermore, the importance of hybrid spatial models lies in charting

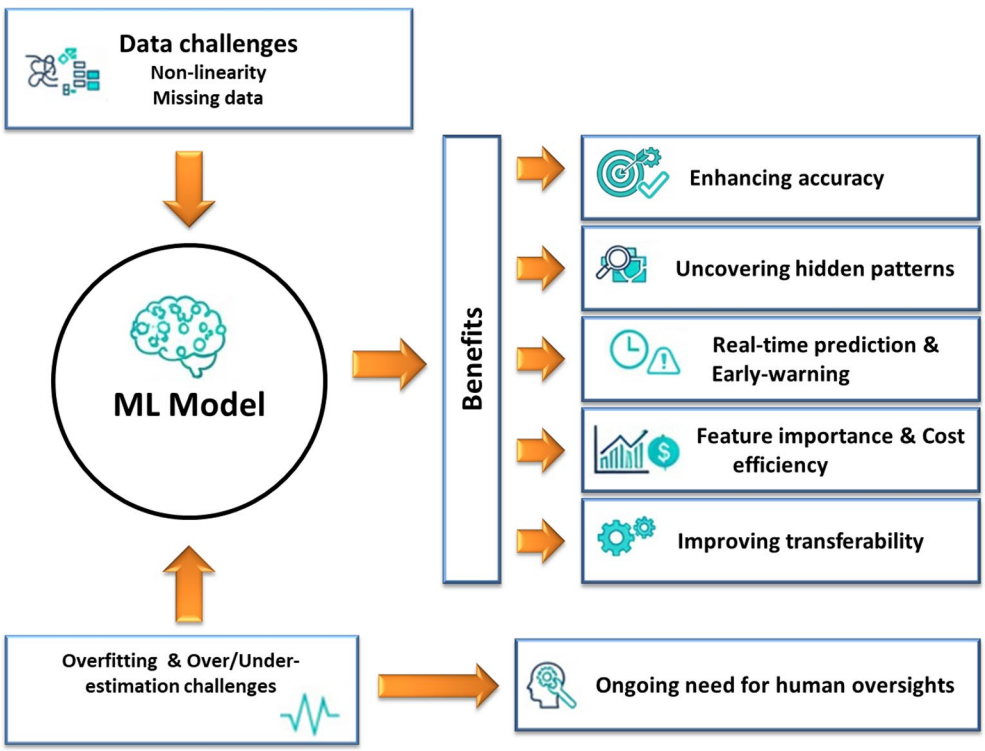


Fig. 1 Machine learning functions and challenges flowchart

new and sustainable research directions. This review first addresses the challenges of nonlinearity and data scarcity, then moves to strategies for reducing monitoring costs. It further explores applications in real-time prediction and early warning systems, alongside model transfer to unmonitored areas. Additionally, the study examines unsupervised learning for detecting hidden patterns without prior labeling. Finally, the synthesis evaluates technical risks, such as overfitting and estimation challenges, to ensure model reliability. The study emphasizes the need for continuous human oversight alongside automation to ensure the reliability of models. Moreover, the trade-off between model complexity and interpretability provides a nuanced understanding of the field, ensuring the reliability of predictions.

Methodology

This systematic review followed PRISMA guidelines to ensure accuracy and transparency (Fig. 2). The literature search was conducted across Web of Science, Scopus, and Google Scholar databases, covering the period from 2014 to 2026. Search terms included keywords such as machine learning, artificial intelligence, water quality prediction, data scarcity, and feature selection. The selection process followed a three-stage screening: initially, 170 articles were excluded during the title screening phase (duplicates or out of scope). Subsequently, 85 articles were removed during the abstract review phase for lacking specific machine learning applications. Finally, 39 articles were excluded after full-text assessment because they did not provide standardized operational performance metrics (e.g., R^2, RMSE), which are essential for the comparative analysis in this review.

Following this rigorous screening, 86 peer-reviewed articles from high-impact journals (Q1/Q2) were selected for final analysis. Each study was scored based on data source reliability, methodology clarity, and the statistical significance of the reported results. These findings are categorized into functional themes, including: (1) Handling complex and missing data, (2) Feature importance and dimensionality reduction, (3) Predictive modeling for early warning, and (4) Regional management and spatial transfer. This approach enables a critical appraisal of how machine learning practically overcomes operational challenges in water quality management. The results relating to model superiority were systematically analyzed by placing the performance of each algorithm in its specific hydrogeological context and data constraints, which were discussed extensively in sections "No one-size model fits all scenarios" and "Why ML performance differs across contexts".

Handling nonlinearity in datasets

Water systems exhibit complex, nonlinear interrelationships driven by geochemical factors, land use, and climate dynamics. Traditional linear models often fail to capture these fluctuations. Machine learning addresses

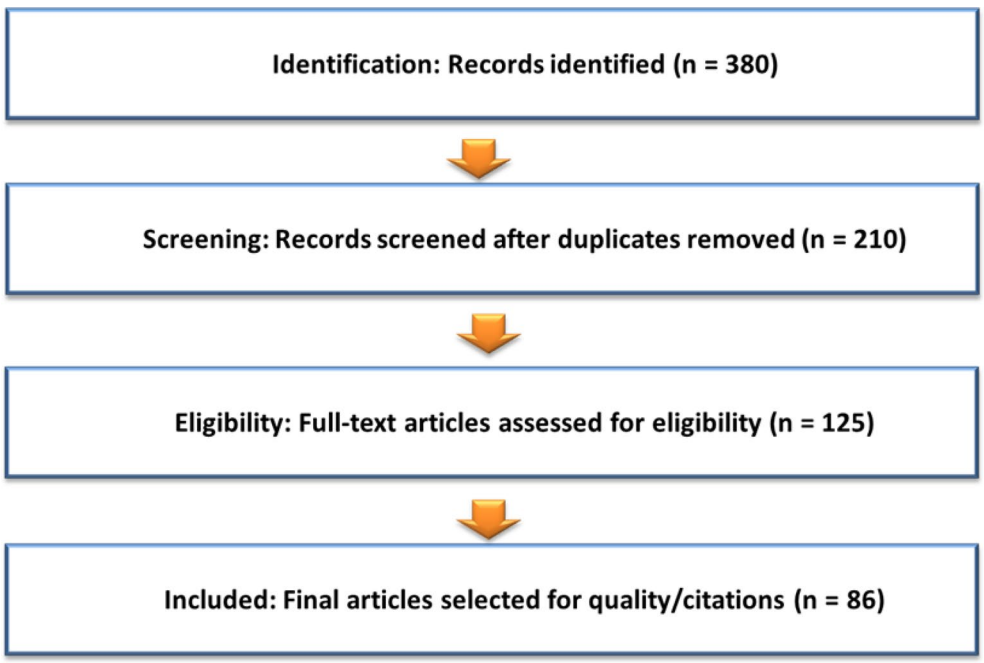


Fig. 2 PRISMA flow diagram describing the selection process

these limitations by identifying complex patterns beyond the scope of linear equations. Consequently, advanced models such as random forest (RF) and artificial neural networks (ANN) have outperformed traditional approaches within specific datasets and non-linear conditions. For example, Castrillo and García [3] and Nacar et al. [25] attributed the accuracy of RF and ANN models to their high generalizability. Their findings indicate that these models significantly outperformed linear methods. This superiority was particularly evident in predicting nutrients, dissolved oxygen (DO), and biological oxygen demand (BOD) [2]. This superiority extends to variable selection, where ML detects non-linear dependencies that statistical tests often overlook. For instance, Chatterjee et al. [7] demonstrated that while statistical tests reduce variables to five, ANN models refine them to a single significant variable without losing accuracy. Similarly, ensemble methods and deep learning models offer robust alternatives for river water quality management. These approaches model internal data complexities with significantly higher reliability than linear regressions [26, 27].

However, complexity does not make linear models outdated. Nonlinear methods carry higher overfitting risks, especially when data relationships are quasi-linear [28, 29]. In these cases, simpler models like ridge regression offer superior accuracy and interpretability. This imposes a trade-off between complexity and generalization. While over-parameterized models can generalize well [30], hybrid frameworks often provide the optimal solution. Recent studies used hybrid ensembles to effectively manage data instability [17, 31]. Consequently, while linear models remain fundamental, machine learning provides the flexibility to map complex aquatic cause-and-effect relationships [24, 32].

Managing poorer or missing data

Incomplete datasets pose a major challenge in water quality monitoring. Data gaps often arise from sensor failures or inconsistent sampling protocols. ML models address this problem not just by ignoring missing values, but by reconstructing underlying patterns. Barcala et al. [12] and Rodríguez et al. [11] demonstrated that RF and SVR efficiently fill data gaps as high as 70%. The success of these models is due to exploiting non-linear correlations between variables to accurately estimate missing values. This ability is a crucial advantage lacking in traditional interpolation methods. Therefore, ML acts as a strategic bridge to overcome monitoring gaps in data-poor areas.

Beyond simple imputation, handling noise and outliers is critical for model stability. Hybrid approaches have proven particularly effective in this regard. For instance, Lu and Ma [33] integrated CEEMDAN denoising with ML models. This hybrid approach filters out data noise, significantly improving prediction accuracy compared to standalone models. Similarly, Wu and Wang [34] used wavelet transform with LSTM to address time-series gaps. These studies collectively suggest that preprocessing and hybrid architectures outperform direct modeling when data quality is low.

Furthermore, outlier capping preserves data size better than deletion. As a result, Ahmed et al. [35] and Trabelsi and Bel Hadj Ali [36] improved accuracy by capping outliers rather than deleting them. This approach prevents the loss of valuable information in small datasets. Recent reviews [37, 38] highlight machine learning as a flexible framework for water quality management. These models effectively compensate for missing values by considering spatiotemporal dimensions. This feature is essential for managing groundwater systems, especially in data-poor regions.

Feature importance and cost efficiency

Reducing the number of input variables is a strategic approach to minimize the time and cost of water quality management. Machine learning (ML) models excel at identifying significant variables while maintaining high predictive accuracy. Studies by Toumi et al. [4] and Ahmed et al. [35] emphasize the importance of easily measurable parameters, such as pH and EC. Prioritizing variables that are easy to measure enables cost-effective and real-time monitoring. Traditional statistical tests often identify numerous significant factors, while machine learning achieves high accuracy with fewer variables. Chatterjee et al. [7] demonstrated this by reducing five significant variables to a single variable (Cr) without compromising model performance.

Model sensitivity to specific variables varies based on the algorithm and the nature of the water system. Kim et al. [39] found that LSTM models are more robust than SVR or RF when difficult-to-measure parameters (like COD) are removed. Similarly, feature selection techniques help identify the influencing factors for each field. For example, surface water quality is primarily driven by organic indicators (DO, BOD, and COD), whereas groundwater remains more sensitive to dissolved minerals like Mn and Cl^- (Table 1; [40, 41]). This distinction enables decision-makers to prioritize critical indicators, thereby accelerating assessments and improving monitoring reliability [42, 43]. Therefore, ML acts as a strategic tool for improving monitoring networks in resource-constrained environments.

No one-size model fits all scenarios

The comparative data in Table 1 indicates that model performance is highly dependent on the problem nature, sample size, and environmental conditions. Consequently, no single model is absolutely superior. This is explicitly demonstrated by Rahman et al. [47], where the optimal model varied entirely based on the dataset: RF performed best for CCME data, Linear regression (LR) for Brown data, and XGBoost for SRDD data. Furthermore, algorithmic hierarchy shifts across different hydrogeological contexts, while SVM models outperformed ANNs in certain studies [45], ensemble methods like gradient boosting and RF have proven more efficient in others [46, 49]. This variability is often related to the model's architecture. For example, ANNs may be sensitive to data noise, while SVMs typically exhibit higher resistance [50]. The data further reveals that while massive datasets (e.g., 33,612 and 29,159) achieve near-perfect accuracy (R^2: 0.97-1.00) via K-fold validation, high performance (R^2: 0.99) is still attainable with smaller samples (e.g., 360 records in [27]). Conversely, larger datasets do not always guarantee superior precision, as evidenced by [26], where 5058 records yielded a comparatively lower accuracy(R^2: 0.69-0.82). This underscores that model reliability is governed by environmental complexity and validation rigor rather than sample size alone. Therefore, selecting an optimal model requires a comprehensive evaluation of data complexity and specific prediction objectives.

Real-time prediction and early-warning management

The integration of ML with Internet of Things (IoT) frameworks has revolutionized dynamic monitoring and early-warning systems. Real-time data transmission enables immediate adjustments and the detection of anomalies, which is crucial for sustainable water management. Advanced systems such as WatAI and HydroVision demonstrate the ability to process high-frequency flow data and RGB images of the Earth's surface [14, 15]. This processing enables the immediate detection of pollution trends with high accuracy. Thus, these technologies help bridge the gap between data collection and field operational response.

Recent advancements highlight the role of hybrid models and ensemble learning in enhancing reliability. Shahid et al. [51] and El-Shafeiy et al. [52] achieved accuracies exceeding 92% in detecting water quality anomalies using IoT-integrated ML. By focusing on fast-response sensors and excluding non-real-time criteria such as BOD/COD, these frameworks provide proactive support for decision-making [31]. Collectively, these applications transform machine learning from a theoretical tool into a practical operational necessity. They enable the precise identification of high-risk areas, facilitating the immediate implementation of necessary preventive measures [53, 54].

Improving transferability through hybrid and spatial models

A key advantage of hybrid AI-geospatial models is transfer learning, which leverages knowledge from data-rich to data-poor regions [55]. By integrating machine learning with geostatistics, these systems significantly enhance predictive power and regional generalization. Such frameworks provide the high spatial accuracy required for monitoring heterogeneous aquifers. For instance, Wunsch et al. [56] used self-organizing maps to classify hydrological patterns, revealing hidden behaviors such as seasonal fluctuations and pumping effects. This approach has proven highly transferable to different time-series applications.

Recent studies demonstrate the effectiveness of integrating AI with GIS and IoT. Sudasman and Gereuw [16] achieved accuracy exceeding 90% in spatial risk analysis. Phong et al. [57] demonstrated the superiority of the MRAB-FT model in efficiently identifying groundwater potential. Furthermore, hybrid ML–geostatistical approaches enable the precise identification of pollution sources [58–60]. Specifically, [58] found that RF and SVM outperformed ANN in predicting heavy metal concentrations by linking them to urbanization factors.

Transfer learning also improves model robustness across different watersheds with minimal recalibration [61, 62]. Peng et al. [63] proposed a transformer-based TLT model for knowledge transfer between monitoring stations, improving pH, oxygen, and nitrogen predictions while avoiding overfitting. Similarly, Luo et al. [64] developed LSTM and BP models by exploiting upstream and downstream correlations, increasing Nash–Sutcliffe (NSE) efficiency by up to 17%. Raj et al. [65] also presented a hybrid convolutional graph LSTM model to enhance prediction accuracy. In a related vein, the DeepSalt framework by Dey et al. [66] demonstrated the ability of deep learning to bridge the gap between laboratory measurements and satellite data.

Additional applications include the HydroVision approach [15], which estimates water quality using standard RGB images. Wang et al. [67] also developed a case-based inference approach to achieve regional generalization of models. In recent reviews, Zhi et al. [68] and Mehrabi et al. [69] noted that algorithms such as RF and XGBoost demonstrate high accuracy. However, challenges such as geographic noise and model interpretability remain and require further investigation. Together, these integrated frameworks support regional mapping and policymaking, particularly in data-scarce regions.

Table 1 Key variables and machine learning models performance in water quality prediction studies (N/A: Not reported)

Study	Model used	Sample size	Inputs	Model ranking	Accuracy/R^2	Validation method	Key features	Reference
1	ANN, Multiple NN	N/A	pH, DO, SS, NH_3-N, while COD and BOD were excluded because they are not measured in real time	Multiple NN > ANN	0.90–0.93	Data splitting (70% training, 15% testing, 15% validation)	All selected parameters	[31]
2	GEP, ANN, LRM	360	TDS, pH, Ec, major ions	GEP > ANN > LRM	0.88–0.99	Data splitting (70% training, 30% testing,)	HCO_3^-, SO_4^{2-}, Cl^-	[27]
3	ANN, SVM	N/A	River flow, temperature, rainfall, flow travel time, DO, TN, and TP	SVM > ANN	0.7938–0.9366	Chronological multi-scenario splitting		[44],
4	ANN, SVM, GMDH	N/A	pH, EC, Temp,TDS, HCO_3^-, SO_4^{2-}, Cl^-, Na^+, Mg^{2+}, and Ca^{2+}	SVM > ANN > GMDH		Data splitting: 70–80% training, 20–30% testing		[45]
5	Several ensemble/standalones models	33,612	pH, DO, CODMn, and NH_3-N	DT, RF, DCF most performance	0.97–0.99	Hold-out (80:20) with k-fold (k = 5)	pH, DO, CODMn, NH_3-N	[46]
6	Decision Trees, Random Forest, and Xgboost algorithms	29 datasets	pH, NH_3-N, Mn, Ni, B, Pb, Zn, F^-, COD, Fe	Xgboost > Decision Trees > Random Fore	Xgboost: 100% (Best performance/No misclassifications) Decision Tree: 100% (High precision/No misclassifications) Random Forest: 96.5% (One false negative case)	Data splitting (70% training, 30% testing,)	pH, Mn, Ni	[40]
7	WDT-ANFIS, ANFIS	field data from 2009 to 2010	12 input parameters	WDT-ANFIS > ANFIS	0.85–0.96	Cross-validation to reduce RMSE	_	[32]
8	Random forest (RF), support vector regression (SVR), artificial neural networks (ANN), and adaptive boosting (AdaBoost)	72	EC, pH, T	AdaBoost > ANN > RF	0.65–0.89	Data splitting (80% training, 20% validation) with fivefold cross-validation	EC > pH > T	[36]
9	Ensemble ML-WQI model, linear regression (LR), random forest (RF), and extreme gradient boosting (XGB) as base-estimators, decision tree (DT)	29,159	NH_3, BOD, ALk, Cl^-, EC, DO, PO_4^{3-}, pH, T, TH	RF was best for CCME data, LR was best for Brown data, and XGB was best for SRDD data	$R^2 = 1.00$	Training dataset (80%) and testing dataset (20%) with fivefold CV		[47]

Table 1 (continued)

Study	Model used	Sample size	Inputs	Model ranking	Accuracy/R^2	Validation method	Key features	Reference
10	ANN, MLR	N/A	pH, T, TSS, TS	ANN > MLR	0.3–0.83	Data splitting (80% training, 20% validation)	pH	[2]
11	SVM, MLP, RF, XGBoost, LSTM	5058	NH_3-N, T, pH, DO, $KMnO_4$, Total TP, TN, EC, and Turbidity	LSTM > SVM > MLP > RF > XGBoost	0.69–0.82	Training dataset (90%) and testing dataset (10%)	EC	[26]
12	RF, M5P, RT, REPT, and 12 hybrid algorithms	685		Hybrid BA-RT outperformed the other models	$R^2 = 0.941$, RMSE = 2.71	Data splitting (70:30) with tenfold CV	Fecal coliform (FC)	[48]
13	Decision Tree (DT), ANN, K-Nearest Neighbors (KNN), Naïve Bayes (NB), Support Vector Machine (SVM), Random Forest (RF), Gradient Boosting (GB)	N/A	13 chemical variables (e.g., DO, BOD, NH_3-N, pH, TSS, COD, temperature, turbidity, conductivity, salinity, nitrogen, phosphorus, E. coli)	Gradient Boosting (learning rate 0.1) with accuracy ≈94.90%, followed by Random Forest (93.88%)	93.88–94.90%	tenfold cross-validation with a 70:30 training/testing ratio	TSS > NH_3-N > BOD > COD > Turbidity > DO	[49]

Uncovering hidden patterns in datasets without prior labeling

Uncovering hidden patterns is a fundamental capability of machine learning, enabling the identification of pollution sources without prior labeling. By analyzing complex relationships, models can detect distinctive chemical signatures from agriculture, wastewater, industrial waste, and traffic. For instance, Lee et al. [13] combined self-organization mapping (SOM) and fuzzy clustering (FCM) to classify groundwater in Seoul into three distinct groups: natural factors, sewage leaks, and subway-related nitrate pollution. Similarly, Yao et al. [70] used SOM and FCM in the Sichuan basin to distinguish between anthropogenic sources (nitrates/ammonium) and natural water–rock interactions (TDS/major ions).

ML models also prevent mistaken conclusions by identifying the true sources of pollution. For example, Tran et al. [24] demonstrated that vertical hydraulic conductivity, rather than seawater intrusion, controls salinity in some coastal areas. Furthermore, Wunsch et al. [56] applied SOM-ensembles to group hydrographs into dynamically similar assemblies, revealing seasonal and pumping effects that were ignored by traditional methods. In urban China, Zhang et al. [58] used RF, ANN, and SVM to separate heavy metals into human-influenced (Zn, Pb, Cu) versus naturally occurring (As, Ni, Cr) elements.

The power of blind machine learning approaches is evident in their ability to decompose chemical mixtures without prior source knowledge. Shaughnessy et al. [71] and Vesselinov et al. [72] used non-negative matrix/tensor factor analysis (NMF/NTF) to automatically identify flow paths and geochemical fingerprints. Additionally, Szomolányi and Clement [73] emphasized the distinction between detecting inherent patterns in unsupervised models and predictive patterns in supervised models, such as random forests.

Distinguishing between natural and anthropogenic influences remains a critical application. Ransom et al. [74] identified well depth, fertilization and climate as key predictors of nitrate in groundwater. This aligns with findings by Tran et al. [24] on groundwater salinity, where models separated natural geochemical signals from over-pumping effects. Advanced architectures, such as the sub-model integration by Wen et al. [17], allow for capturing complex relationships and detecting anomalous methane associated with anthropogenic activities. In soil-crop systems, Hu et al. [60] found that plant type and organic matter are key factors in heavy metal accumulation. These findings support the studies of Zhang et al. [58] and Shi et al. [59], whose spatial frameworks mapped the distribution of heavy metals and prioritized influencing factors. Collectively, these studies confirm ML as a flexible and accurate tool for understanding the mechanisms that control water quality in complex ecosystems.

Challenges of over-fitting and over-estimating/under-estimating in machine learning models

Over-fitting and estimation biases (over/underestimation) are major challenges affecting the accuracy and reliability of ML models in water quality prediction. Over-fitting occurs when a model learns noise and site-specific details instead of generalizing patterns. This results in high training performance but poor generalization to new data [23, 75, 76]. An unusually high R^2 (near 1) or low MSE (near 0) may be problematic, indicating a strong possibility of overfitting (https://statisticsbyjim.com/regression/r-squared-too-high/). Similarly, Palabıyık and Akkan [77] suggested that linear relationships in water parameters may mask overfitting problems.

To mitigate over-fitting, Ying [23] suggested strategies such as early-stopping, network reduction, data expansion, and regularization. For instance, Xiao et al. [78] used Particle Swarm Optimization (PSO) to fine-tune BPNN parameters, preventing the model from merely memorizing training data. Similarly, Bouihi et al. [75] successfully applied L1 (Lasso) and L2 (Ridge) regularization to balance complexity and data size, achieving 89% accuracy. In an unconventional view Belkin et al. [30], introduced the double descent theory to explain the performance of complex models. The theory suggests that over-parameterized models can transcend the classical U-curve of the bias-variance trade-off. This is achieved by selecting smoother and simpler solutions, through inductive bias beyond the interpolation threshold.

Beyond overfitting issues, models often suffer from overestimation or underestimation biases. Overestimation creates a falsely optimistic view of water quality, which may mask environmental risks. Conversely, underestimation predicting poorer quality than reality is often considered a safer bias for environmental decision-making. This was highlighted by Zheng et al. [40] who showed that the SWM model recorded 0% overestimation but 34% underestimation, encouraging a cautious approach. Furthermore, Babitsch and Sundermann [79] highlighted that small, right-skewed datasets often trigger these biases, necessitating larger samples and improved monitoring designs. To address these persistent challenges, Rahman et al. [47] proposed ensemble ML-WQI models to enhance generalization and reduce estimation errors in unbalanced datasets. In this context, as shown in Table 1, and to ensure the robustness of the results, K-fold cross-validation relies on rotating the data (e.g., $K=5$ means splitting the data into 5 parts and testing each part alternately) to reduce bias (RMSE) and improve model reliability.

Table 2 ML functions and challenges in water quality management

Key themes	Key findings	Key algorithms	Reference
Handling nonlinearity	ML models significantly outperform linear models in complex datasets. For instance, RF reduced RMSE by 60.1% compared to linear models, and ANN showed superior predictive capability for BOD/COD over MLR	RF, ANN, LSTM, Deep Learning	[2, 3, 8, 9]
Data scarcity and imputation	ML models successfully reconstruct time-series data with 50–70% missing data (NSE > 0.45). Hybrid denoising models improve stability in data-poor environments	RF, SVR, CEEMDAN-RF, LSTM	[11, 33]
Feature optimization (cost reduction)	Reducing the number of required monitoring parameters (e.g., from 13 to 4 variables) maintains accuracy while lowering costs. pH, EC, DO, and NH_3-N have consistently been identified as critical parameters	Feature selection, sensitivity analysis	[26, 27, 40, 85]
Real-time monitoring and transferability	Integration with IoT achieved up to 99.8% anomaly detection accuracy. Transfer learning improved prediction efficiency (NSE + 7–17%) in unmeasured regions without extensive recalibration	IoT-Ensembles, Transfer Learning, Geospatial AI	[16, 51, 57, 64]
Pattern recognition and source tracking	Unsupervised learning effectively identifies pollution sources (e.g., distinguishing between sewage and natural rock-water interactions) and groundwater salinization factors without prior classification	SOM, Fuzzy Clustering, Unsupervised Learning	[13, 24]
Challenges and limitations	High risk of overfitting in small datasets (high training R^2 vs. low testing performance). Automated workflows (AutoML) still need human oversight for validation and interpretation	Regularization (L1/L2), early stopping, human-in-the-loop (HITL)	[23, 75, 80, 81]

Beyond automation: Ongoing need for human oversight in machine learning models

AutoML systems effectively accelerate data science workflows. These technologies enable machine learning applications even for researchers who do not have in-depth computational backgrounds [80]. However, while these systems automate numerous steps, they do not eliminate the critical role of human expertise in design and tuning. Studies by Xin et al. [81] emphasize that automation should be a supportive partner, balancing simplicity, reproducibility and reliability. Furthermore, Crisan and Fiore-Gartland [80] highlight that human intervention remains essential for interpreting results within a specific application context and for making informed decisions.

Practical applications often require manual preprocessing to ensure data consistency. For instance, Shams et al. [82] demonstrated that handling missing and outliers values, and standardizing metrics, followed by grid search optimization, is critical for accurate water quality prediction. Although Mumuni and Mumuni [83] reported significant progress in automating tasks such as data cleaning and feature selection. However, they emphasized that complex data still requires human intervention at various stages of development.

The architecture of the data pipeline itself is a crucial factor. Quemy [84] showed that the design of the processing stage often has a greater impact on model performance than mere hyperparameter tuning. This aligns with findings by Tran et al. [24], who noted that accuracy depends on the synergy between the chosen algorithm, prediction goals, and input variables. Collectively, these studies indicate that machine learning systems operate most effectively within human-led frameworks, ensuring that the methodology remains robust and contextually relevant.

Results and discussion: From prediction to management

Building upon previous sections, this section synthesizes the operational gains of ML models, moving from technical description to a comparative interpretation of their efficiency in water quality management. This review scans the performance of machine learning models in handling complex environmental data, addressing data scarcity, and reducing monitoring costs. The results demonstrate the specific advantages of ML models over traditional approaches in terms of accuracy and flexibility. Table 2 summarizes the core functions of ML models in

water quality management, highlighting the most important findings and existing challenges.

Model complexity vs. applicability

Published studies show that water quality data are inherently complex and subject to nonlinear relationships. Traditional linear models often fail to capture these dynamics. Consequently, machine learning techniques have demonstrated robust efficacy in modeling these complex systems, though their outperformance remains strictly dataset-dependent. This superiority is quantified in this review through specific operational metrics rather than general accuracy claims. For example, the RF model was 60.1% more efficient at reducing the RMSE value compared to linear models when predicting nutrients [3]. Similarly, neural networks (ANNs) outperformed multiple linear regression (MLR) in estimating BOD and COD due to their effectiveness in processing complex, high-dimensional data [2]. Furthermore, deep learning models like LSTM have proven effective in modeling time-dependent dynamic fluctuations [8, 9]. While ML dominance is well-established, it is important to acknowledge nuanced exceptions where complexity does not guarantee superiority. Simple linear models may outperform complex techniques when data relationships are quasi-linear or when interpretation is prioritized [28, 29]. Therefore, the optimal choice of model should be context-dependent rather than a one-size-fits-all approach. This balanced view ensures that model selection aligns with both data characteristics and management goals.

Data scarcity and reliability of imputation

A key finding of this review is the remarkable ability of machine learning models to operate in data-poor environments. RF and SVR techniques have proven highly effective in addressing time series gaps. They have maintained acceptable performance (NSE > 0.45) even with 50–70% data loss [11]. Hybrid models, such as CEEMDAN-RF, employ noise reduction techniques to improve prediction stability even with sparse or noisy data [33]. This capability enables the recovery of basic patterns without costly field measurements. The matter makes machine learning an effective tool for water management in data-poor developing regions.

Economic feasibility and dimensionality reduction

The reviewed studies indicate that ML models can significantly reduce monitoring costs by focusing on a limited set of key variables. This is known as feature significance analysis. For example, studies have shown that WQI prediction can be simplified from 13 parameters to just 4 (such as BOD, NH3, FCB, and pH) without compromising accuracy [85]. By reducing the required variables from 13 to 4, ML provides a cost-effective framework for sustainable water management. As summarized in Table 1, parameters such as pH, EC, DO, and ammonia (NH_3-N) consistently appear as the most influential variables across various studies [26, 27, 40]. By focusing on these key indicators, decision-makers can utilize cost-effective, real-time monitoring systems. This reduction in dimensions not only saves costs but also improves model generalizability by excluding irrelevant or redundant features.

Real-time monitoring and transferability

The integration of ML with IoT and geospatial technologies has revolutionized early-warning systems. Deep learning frameworks and IoT-integrated ensembles (e.g., RF, XGBoost) achieved an accuracy of 99.8% in detecting anomalies and classifying water quality in real time [51]. Moreover, the review highlights the emergence of transfer learning as a solution for spatial generalization. This approach allows models trained in data-rich regions to be adapted to data-poor regions. This significantly improves forecasting efficiency (NSE increasing by 7–17%) without the need for extensive recalibration [64]. Hybrid AI-geospatial models further enhance this by mapping spatial hazards and groundwater potential with high accuracy [16, 57].

Diagnostic insights for controlling pollution sources

Beyond prediction, ML models are powerful diagnostic tools. Unsupervised learning techniques have successfully identified pollution sources without prior labeling. These models can distinguish natural geochemical influences from anthropogenic activities. For example, self-organization maps (SOMs) and fuzzy clustering have successfully differentiated between sewage seepage and natural rock-water interactions [13]. They have also demonstrated similar efficiency in identifying groundwater salinity in coastal areas, attributing it to upconing process rather than seawater intrusion [24]. This ability to automatically analyze chemical mixtures into their source components offers a significant advantage over traditional inverse modeling.

Human-AI synergy: overcoming automation limitations

Despite these progresses, the application of ML in water quality management still faces challenges, most notably overfitting and estimation bias. High R^2 values in training coupled with poor testing performance often indicate overfitting, particularly in small or noisy datasets [23, 75]. Strategies such as regularization (L1/L2), early stopping, and ensemble modeling have been

identified as effective mitigation measures [75]. Furthermore, studies warn against relying entirely on automation, as human oversight remains essential for cleaning up data, outlier detection, and model interpretation. Automated machine learning (AutoML) should be viewed as a supportive tool, not a replacement for specialized expertise in the field [80, 81].

Why ML performance differs across contexts

The review confirms that the superiority of one model over another is not merely an algorithmic difference but depends entirely on the context. This variability is related to the nature of the aquatic system, data quality, and the degree of human intervention in model tuning. Therefore, there is no single model suitable for all global water quality management situations. In groundwater systems with predictable recharge, simple models such as linear regression can perform adequately. However, environments with spatial heterogeneity or industrial pollution require nonlinear models such RF. These advanced models enable the monitoring of sudden shifts in quality indicators with high accuracy. On the other hand, the performance of machine learning is primarily affected by data availability and quality. In data-poor environments, RF and k-NN models excel due to their flexibility in handling missing values. Conversely, deep learning models such as LSTM are known to be data-hungry. In cases of small or sparse data, these models are prone to overfitting [86]. Finally, the efficiency of models varies according to the prevailing variables in local locations. Coastal models may fail in agricultural areas due to the different nature of the dominant pollutants. For example, a model focusing on TDS and EC might work perfectly in coastal aquifers but fail in agricultural zones, where nitrates and pesticides are the dominant nonlinear drivers. This explains why transferable models often require local fine-tuning to maintain accuracy across different basins.

Future research

Future research directions should focus on several key points, most notably:

- Developing frameworks applicable on a broad regional scale.
- Linking ML models to IoT-enabled monitoring systems to provide dynamic feedback and enhance early-warning management.
- Using ML techniques carefully to avoid overfitting or overestimation of their results.
- These advancements will ensure that ML tools move from black-box predictions to transparent management solutions.

Conclusion

Machine learning has become a cornerstone of water science. This research paper reviews the high efficiency of ML techniques in processing nonlinear and missing data. These models are highly effective at prioritizing features importance and enabling real-time spatial predictions. This deepens our understanding of hydrogeological processes, paving the way for smart water-quality management. Integrating machine learning into water quality management represents a shift from reactive assessments to proactive predicting. This integration enhances sustainability and provides real operational gains. One promising future direction is to continue developing the hybrid frameworks discussed previously, integrating machine learning with traditional physical, hydrochemical, and geostatistical models. These hybrid approaches will improve the generalizability of the model, particularly in data-poor regions. They will also enhance the physical interpretability of automated water quality assessments. This integration effectively bridges the gap between data-driven efficiency and physical reliability.

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Declarations

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